Quantification and Relational Division in Double-Categorical OLOGs

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Abstract

Working draft of a paper on quantification, modality, data queries and comprehension, all in the context of double categories. A follow-up to [LP25].

1 Introduction

2 Relational Division, Quantification, and Modality

Relational division is a query introduced in [Cod72, §2] that behaves essentially as a universal quantifier over given specified values in the column of a table. First we illustrate this operation with a very simple generic example. We may as well work with binary relations $R\colon C \nrightarrow A$ which are of course equivalently monic arrows $\langle d, c\rangle\colon R \rightarrowtail C \times A$. Recall the notation from the reference

$$r[A] = c(r) \text{ where } r \in R$$

and likewise for $r[C] = d(r)$. This is a bit notationally-heavy, but is consistent with the reference and allows for generalizations beyond the following example. Likewise, for any binary relation T, let $g_T(x) = \{y \mid xTy\}$, that is, the set of all y in the target of T related to x under T. Call this the T-orbit of x. Now, for the example, consider the two relational tables R and S with columns (C, A) and (B, D), respectively:

R	
C	**A**
1	a
1	b
1	c
2	a
2	b
3	b

S	
B	**D**
a	2
b	1
b	2
a	3

Thus, for example, $2Ra$ and $2Rb$. Likewise, $aS2$, et cetera. Now, in the notation of the reference, a division query is

$$R[A \div B]S = \{r[C] \mid r \in R \wedge S[B] \subset g_R(r[C])\} = \{1, 2\}$$

We unpack this carefully to illustrate the operation. Note that the columns A and B must take the same types of values although the values do not need to have the same exact extension. Now, $r[C]$ is the set of values of the R-column, C. So, $r[C] = \{1, 2, 3\}$. Likewise, $S[B] = \{a, b\}$. So, there are three sets of the form $g_R(r[C])$, only two of which contain $S[B]$, as in:

$$\begin{aligned} S[B] &= \{a,b\} \not\subseteq \{b\} = g_R(3) \\ S[B] &= \{a,b\} \subset \{a,b,c\} = g_R(1) \\ S[B] &= \{a,b\} = \{a,b\} = g_R(2). \end{aligned}$$

Hence again, the division query returns $\{1,2\}$, exactly *the set of all C-values whose orbit contains all the values of the B-column of S*. More prosaically, a C-value is selected by the query iff it is related to everything on the list of B-values. The universal quantification is thus over the set of B-values. The query allows the comparison of separate tables; note that D was not needed to do the query. In the containment relations above, the universally quantified set does not need to completely describe the orbit of any given element satisfying the division query; that is, 1 is related to c, but this is not a B-value. Likewise 3 is related to something on the list, but not everything.

A more concrete example illustrating the same query is the following. Consider the two relational tables below. On the left, rows pair an apothecary with an ingredient sold there; on the right, rows pair an ingredient with an effect produced by the combination of that ingredient with another in a potion.

Apothecary	
Name	**Ingredient**
Arcadia's Cauldron	giant's toe
Arcadia's Cauldron	wheat
Arcadia's Cauldron	void salts
Elgrim's Elixers	giant's toe
Elgrim's Elixers	wheat
The White Phial	wheat

Potion	
Ingredient	**Effect**
giant's toe	fortify health
wheat	fortify health
wheat	lingering damage magicka
giant's toe	damage stamina regen

Note that the second table does not list the other ingredients required to create the potion producing the effect. Our expertise is that any two such ingredients with the same effect will create the desired potion when combined. Now, suppose we want to create a fortify health potion but are to lazy to fight a giant to get his toe. We'd also really like to one-stop-shop (maybe the merchant will give us a deal if we buy in bulk). The orbits are then the inventories:

$$\begin{aligned} g_{\text{Apoth}}(\text{Arcadia's}) &= \{\text{giant's toe, wheat, void salts}\} \\ g_{\text{Apoth}}(\text{Elgrim's}) &= \{\text{giant's toe, wheat}\} \\ g_{\text{Apoth}}(\text{White Phial}) &= \{\text{wheat}\} \end{aligned}$$

The the set of values quantified over is our shopping list: {giant's toe, wheat}. The query is then

$$\text{Apothecary[Ingredient} \div \text{Ingredient]Potion} = \{\text{Name} \mid \{\text{giant's toe, wheat}\} \subset \text{Inventory(Name)}\}$$

which is just to say that we are looking for the proprietors who have all the ingredients on our shopping list. The result is {Arcadia's, Elgrims}. It does not matter that Arcadia's also has void salts; but it does matter that The White Phial only has wheat but no giant's toe.

There is something awkward about these examples inasmuch as we would probably more likely have tables better reflecting our alchemical expertise such as

Apothecary	
Name	**Ingredient**
Arcadia's Cauldron	giant's toe
Arcadia's Cauldron	wheat
Arcadia's Cauldron	void salts
Elgrim's Elixers	giant's toe
Elgrim's Elixers	wheat
The White Phial	wheat

Potion	
Ingredient	**Effect**
giant's toe	fortify health
wheat	fortify health
wheat	lingering damage magicka
giant's toe	damage stamina regen
swamp fungal pod	restore health
void essence	restore health

.

However, division with quantification over the whole the Ingredient column in the Potion table now produces the empty set since no merchant has all those items on hand. And [Cod72, §2.4] recognizes this in the example queries. Namely, example query #8 to find suppliers supplying at least parts 12, 13 and 15 restricts the part # values to only 12, 13 and 15 without ranging over the whole column before executing the division query. This is exactly a *shopping list* kind of example. In other words, technically, the division query is not set up to do selection from a sub-list of a column on its own. A restriction to just the values of interest from the second table is employed implicitly before doing the division query. In the example immediately above, to execute our query, we would filter the rows with the value "fortify health" first and then do the division query with the ingredients column. The flexibility of the division query as written is that it can be applied to any two tables with columns having the same types of values; the expense is that it might not return anything and may require restrictions to produce results in intuitive cases.

3 OLOGs and Instances

What is going on here from a double-categorical OLOG-perspective [LP25] is that we have two basic relations

$$\boxed{\text{Name}} \xrightarrow{\text{Apothecary}} \boxed{\text{Ingredient}} \qquad \boxed{\text{Ingredient}} \xrightarrow{\text{Potion}} \boxed{\text{Effect}} \;.$$

Suppose that these relations are instanced by the two tables at the end of the previous section, thought of as living in $\mathbb{R}\mathsf{el}$. Now, the OLOG generated will be a cartesian equipment. So, given an individual as a global element of a type, say, Arcadia's: $1 \to \boxed{\text{Name}}$, we can form the restriction

$$\begin{tikzcd}
1 \arrow[r, "\text{Arcadia's Inventory}", "\shortmid"{marking}] \arrow[d, "\text{Arcadia's}"'] \arrow[dr, phantom, "\text{res}"] & \boxed{\text{Ingredient}} \arrow[d] \\
\boxed{\text{Name}} \arrow[r, "\text{Apothecary}"', "\shortmid"{marking}] & \boxed{\text{Ingredient}}
\end{tikzcd}$$

which in the instance is precisely the named merchant's inventory. This would play the role of $g_R(x)$ in the previous section. Likewise, our ingredient list for a fortify health potion is schematized by a restriction:

$$\begin{tikzcd}
\boxed{\text{Ingredient}} \arrow[r, "\text{fortify health potion}", "\shortmid"{marking}] \arrow[d] \arrow[dr, phantom, "\text{res}"] & 1 \arrow[d, "\text{fortify health}"] \\
\boxed{\text{Ingredient}} \arrow[r, "\text{Potion}"', "\shortmid"{marking}] & \boxed{\text{Effect}}
\end{tikzcd}$$

These are both examples of "filter" queries discussed in [LP25]. Now, we are trying ultimately to compare the two restrictions, to see if the ingredients for the fortify health potion are in the various inventories. In the examples above the composites

$$\begin{tikzcd} 1 \arrow[r, "\text{Arcadia's Inventory}", "|" marking] & \boxed{\text{Ingredient}} \arrow[r, "\text{fortify health potion}", "|" marking] & 1 \end{tikzcd}$$

and

$$\begin{tikzcd} \boxed{\text{Name}} \arrow[r, "\text{Apothecary}", "|" marking] & \boxed{\text{Ingredient}} \arrow[r, "\text{fortify health potion}", "|" marking] & 1 \end{tikzcd}$$

are natural to study but neither returns what we want. In the first case, the $\mathbb{R}\mathsf{el}$-composite will return whichever ingredient Arcadia's has on the list; in the second case, the $\mathbb{R}\mathsf{el}$-composite returns whichever merchants have an ingredient on the list. This is a result of the fact that existential quantification over the instance of the middle type is built into composition in $\mathbb{R}\mathsf{el}$.

If we allow ourselves tabulators in the OLOG, much more progress can be made. This allows the creation of types with projections and universal cells, namely,

$$\begin{tikzcd} & \boxed{\text{Arcadia's Inventory}} \arrow[dl] \arrow[dr] \arrow[d, phantom, "\text{tab}"] & \\ 1 \arrow[rr, "\text{Arcadia's Inventory}"', "|" marking] & {} & \boxed{\text{Ingredient}} \end{tikzcd}$$

and then another

$$\begin{tikzcd} & \boxed{\text{ingredients for fortify health potion}} \arrow[dl] \arrow[dr] \arrow[d, phantom, "\text{tab}"] & \\ \boxed{\text{Ingredient}} \arrow[rr, "\text{fortify health potion}"', "|" marking] & {} & 1 \end{tikzcd}.$$

Now, the shopping query is thus to ask whether for each such inventory, the data reflects a putative relationship schematized as

$$\begin{tikzcd} \boxed{\text{ingredient for fortify health potion}} \arrow[rr, dashed] \arrow[dr] & & \boxed{\text{Arcadia's Inventory}} \arrow[dl] \\ & \boxed{\text{Ingredient}} & \end{tikzcd}$$

where, in other words, the inventory in question contains the ingredients on the list. Of course this is not quite right, because we'd like to look at all such inventories simultaneously and pick those merchants whose inventories contain all the ingredients. All the inventories will be contained in our instancing of the ingredient type. So, a better way to do this is just to isolate the type of ingredients and view our shopping list as a subtype of the type of ingredients:

$$\begin{tikzcd} \boxed{\text{ingredient for fortify health potion}} \arrow[d, tail] \\ \boxed{\text{Ingredient}} \end{tikzcd}$$

This doesn't necessarily need to be monic. But monic arrows are perfectly well axiomatizable in our framework. Perhaps we use the notation to indicate we are thinking of it as a subtype obtained by comprehension [Jac99, §9.3]. Nor does it matter how the subtype comes about at this stage; it could have been obtained as a tabulator as above, or simply required as a primitive in the setup of the OLOG.

What matters is comparing this subtype with the inventories of all the merchants simultaneously. These are implicit in the instancing of the Apothecary relation, so we study the corner

$$\begin{tikzcd} & \boxed{\text{ingredient for fortify health potion}} \arrow[d, tail] \\ \boxed{\text{Name}} \arrow[r, "|" marking, "\text{Apothecary}"'] & \boxed{\text{Ingredient}} \end{tikzcd}$$

It is tempting just to restrict with an identity morphism on the left. This results in a cartesian cell

$$\begin{tikzcd} \boxed{\text{Name}} \arrow[r, "|" marking, "\text{Merchant with fortify health ingredient}"] \arrow[d, equal] \arrow[dr, phantom, "\text{res}"] & \boxed{\text{ingredient for fortify health potion}} \arrow[d, tail] \\ \boxed{\text{Name}} \arrow[r, "|" marking, "\text{Apothecary}"'] & \boxed{\text{Ingredient}} \end{tikzcd}$$

which filters the table for those rows having one of the two ingredients on our list:

Apothecary	
Name	**Ingredient**
Arcadia's Cauldron	giant's toe
Arcadia's Cauldron	wheat
Elgrim's Elixers	giant's toe
Elgrim's Elixers	wheat
The White Phial	wheat

Notice that this gets rid of the unnecessary entry with the value "void salts" but it does not establish our query. Another operation with which our OLOG comes equipped is extension. In $\mathbb{R}\mathsf{el}$ this acts as an image, and formalizes a select query [LP25]. In the present instance we might try an extension

$$\begin{tikzcd} \boxed{\text{Name}} \arrow[r, "|" marking, "\text{Merchant with fortify health ingredient}"] \arrow[d, equal] \arrow[dr, phantom, "\text{ext}"] & \boxed{\text{ingredient for fortify health potion}} \arrow[d, "!"] \\ \boxed{\text{Name}} \arrow[r, "|" marking, "\text{Merchant with fortify health ingredient}"'] & 1 \end{tikzcd}$$

which in the data instance is effectively just a list of the three merchants, since each has at least one ingredient. But if we went to the White Phial, we'd be out of luck. This result is owing to the fact that in $\mathbb{R}\mathsf{el}$ the extension is an image computed via *existential quantification.* So, the composite query here of restriction along the subtype, followed by extension has performed an *existential query*, returning each merchant whose inventory contains an ingredient on our list.

Now, this is clearly not the result we want, but it is not a dead-end, for it suggests a path forward. Extensions are essentially left adjoints to restriction. Inasmuch as extension is thought of as existential quantification and restriction is thought of as substitution, we have an analogy of adjunctions

$$\mathrm{ext} \dashv \mathrm{res} \qquad \leftrightarrow \qquad \exists \dashv (-)^*.$$

Now, of course in nice logical situations substitution also has a right adjoint, which is *universal quantification*. This is of course exactly the kind of operation we are seeking to formalize. What we want then is to complete the adjoint analogy

$$\mathrm{ext} \dashv \mathrm{res} \dashv ??? \qquad \leftrightarrow \qquad \exists \dashv (-)^* \dashv \forall$$

in the theater of OLOGs and of double categories generally. Something like this has been studied in the foundational paper [Shu08] under the guise of *coextension*. It was mentioned there, but not studied in-depth due to being not as well behaved as the other two operations. And indeed this is not what we envision in this paper. For coextensions are had by asking for the equipment $\mathbb{D}_1 \to \mathbb{D}_0 \times \mathbb{D}_0$ to be also a *cofibration* which is too strong for what we have in mind. Rather, we take the approach of *fibrational semantics* [Jac99] and ask merely that our equipments $\mathbb{D}_1 \to \mathbb{D}_0 \times \mathbb{D}_0$ have right adjoints to restriction functors and that these satisfy a Beck-Chevalley condition. This is exactly to require *fibered products*. We shall see in more detail how this recovers exactly the queries of interest below in Section 7. For this we first review quantification in more detail.

4 Varieties of Quantification

Both universal and existential quantification occurred in some form in the discussion of Section 3. In general, it is well-known that quantification provides left and right adjoints to substitution. A proposition is a subset $S \subset X$ or perhaps a monic arrow $S \rightarrowtail Y$. Such propositions are in 1-1 correspondence with functions $X \to \mathbf{2}$. That $x \in X$ satisfies the proposition $S \subset X$ is written $S(x)$ and means $x \in S$ too. Now, for a set function $f\colon X \to Y$, there is a usual adjoint situation

$$\begin{tikzcd} PX \arrow[r, shift left=2, "\forall_f"] \arrow[r, shift right=2, "\exists_f"'] & PY \arrow[l, "f^*"] \end{tikzcd} \qquad\qquad \exists_f \dashv f^* \dashv \forall_f$$

where

$$\exists_f(S) = \{y \in Y \mid f(x) = y \text{ for some } x \in S\}$$
$$\forall_f(S) = \{y \in Y \mid \text{ for all } x \in X \text{ if } f(x) = y \text{ then } x \in S\}.$$

Of course ordinary quantification is a special case illustrated by taking $f = \pi\colon X \times Y \to Y$, the usual projection to the second factor of the cartesian product. In this case, the quantified subsets are $R \subset X \times Y$, that is, binary relations. That x and y are R-related could be written xRy or perhaps $x \leq_R y$ or $x \sim_R y$. Quantification is then:

$$\exists_\pi(R) = \{y \in Y \mid R(x, y) \text{ for some } x \in X\}$$

$$\forall_\pi(R) = \{y \in Y \mid R(x, y) \text{ for all } x \in X\}.$$

Typically, the left variable is bound. Note also in particular that if $Y = 1$, we have closed formulas *viz.* truth values.

Now, a relation $R\colon X \nrightarrow Y$ might also be thought of as set of pairs of entities and observations or attribute values, or when $X = Y$ as a non-deterministic system. Fixing x and quantifying over y instead suggests, due to non-symmetry, that y is somehow subsequent to, or a consequence of x.

This comports with the system view of R that y is a later state accessible from x. Perhaps x is prior to y in a temporal sequence. On the causal view, somehow y is a consequence of x. Now, the set of all $x \in X$ for which some subsequent state or attribute value satisfies a proposition $S \subset Y$ is written

$$\Diamond S = \{x \mid S(y) \text{ for some } x \leq_R y\}.$$

This is exactly the modal possibility operator familiar from the topological semantics given by the Alexandrov Topology. Likewise there is the set of those elements of X, all of whose subsequent states or attribute values satisfy $S \subset Y$, written

$$\Box S = \{x \mid S(y) \text{ for all } x \leq_R y\}.$$

This is modal necessity in the same topological semantics. Thus, in the former case, an element of $\Diamond S$ is an element of X for which S is possible, in the sense that it is true of some later accessible state. In the latter case, an element of $\Box S$ is an element of X for which S is necessarily true, in the sense that any subsequent accessible state satisfies S.

So far this is just the usual $\mathbb{R}\mathsf{el}$-semantics of quantification and modality. These operators recover those discussed in Section 3 and several others. If we go back to the table instancing the relation

Apothecary	
Name	**Ingredient**
Arcadia's Cauldron	giant's toe
Arcadia's Cauldron	wheat
Arcadia's Cauldron	void salts
Elgrim's Elixers	giant's toe
Elgrim's Elixers	wheat
The White Phial	wheat

$$\boxed{\text{Name}} \xrightarrow{\text{Apothecary}} \boxed{\text{Ingredient}}$$

and our list S again has wheat and giant's toe, then directly from the operations above we have six queries summarized in Figure 1. Use R to denote the relation and $R^{\dagger}$ is the reverse relation. The variables x and y range over names and over ingredients, respectively. Notice some curiosities. The first is that R is supposed to be given. Thus, the existential queries are just the select column queries returning either all the merchants or all the ingredients specified by R. These are already studied in [LP25]. The universal queries are new and more interesting, but neither returns our list query since S does not even enter into the construction. The modal queries as well are new. The possibility query returns the same as the existential query only as an accident of R. If The White Phial was paired with something other than wheat, for example, the results would be different. Note as well that the necessity query is not the list query we are looking for. It returns the set of those merchants having only the ingredients on our list. This is a query with utility, but as far as our shopping list goes it does not matter whether the returned merchant has other items. In fact, suppose that one closer has our ingredients but also a lot of others. We would naturally prefer to go there.

Result of Existential Query

$$\begin{aligned}\exists_\pi(R) &= \{y \mid R(x,y) \text{ for some } x \in X\}\\ &= \{\text{ingredients someone has}\}\\ &= \{\text{wheat}, \text{giant's toe}, \text{void salts}\}\end{aligned}$$

Result of Universal Query

$$\begin{aligned}\forall_\pi(R) &= \{y \mid R(x,y) \text{ for some } x \in X\}\\ &= \{\text{ingredients everyone has}\}\\ &= \{\text{wheat}\}\end{aligned}$$

Result of Reverse Existential Query

$$\begin{aligned}\exists_\pi(R^\dagger) &= \{x \mid R(x,y) \text{ for some } y \in Y\}\\ &= \{\text{those with an } R\text{-ingredient}\}\\ &= \{\text{wheat}, \text{giant's toe}, \text{void salts}\}\end{aligned}$$

Result of Reverse Universal Query

$$\begin{aligned}\forall_\pi(R^\dagger) &= \{x \mid R(x,y) \text{ for all } y \in Y\}\\ &= \{\text{those with all } R\text{-ingredients}\}\\ &= \{\text{Arcadia's}\}\end{aligned}$$

Result of Possibility Query

$$\begin{aligned}\Diamond S &= \{x \mid S(y) \text{ for some } x \leq_R y\}\\ &= \{x \mid x \text{ has an ingredient on } S\}\\ &= \{\text{Arcadia's}, \text{Elgrim's}, \text{Phial}\}\end{aligned}$$

Results of Necessity Query

$$\begin{aligned}\Box S &= \{x \mid S(y) \text{ for all } x \leq_R y\}\\ &= \{x \mid x \text{ has only ingredients on } S\}\\ &= \{\text{Elgrim's}\}\end{aligned}$$

Figure 1: Results of six quantification queries.

It will turn out that the list query is given by a pair of operations, one of which is universal quantification, applied in sequence. And so, to enrich the datascheme to handle this, the goal now is to formulate all of the above operations purely double-categorically and axiomatize them in our generated OLOGs. Ultimately, we will allow the data-instancing and double-categorical functorial semantics to perform the desired queries.

5 Operations in Double Categories

Example 5.1. In $\mathbb{M}\mathsf{at}(\mathsf{V})$, restrictions and extensions are computed as

$$\begin{tikzcd} Z \arrow[r, "f_! M g^*", "\shortmid"{marking}] \arrow[d, "f"'] \arrow[dr, phantom, "\text{res}"] & W \arrow[d, "g"] \\ X \arrow[r, "M"', "\shortmid"{marking}] & Y \end{tikzcd} \qquad f_! M g^*(z,w) = M(fz, gw)$$

and

$$
\begin{tikzcd}
Z \arrow[r, "N", "|" marking] \arrow[d, "f"'] \arrow[dr, phantom, "\text{ext}"] & W \arrow[d, "g"] \\
X \arrow[r, "f^*Ng_!"', "|" marking] & Y
\end{tikzcd}
\qquad\qquad
f^*Ng_!(x,y) = \sum_{\substack{fz=x \\ gw=y}} A(z,w)
$$

□

6 Existential Queries

The set-up we are entertaining is that of having a subtype $s\colon S \rightarrowtail Y$ thought of as specifying in the associated data some subcollection or list of items in the ambient list instancing Y. We have already seen that in $\mathbb{R}\mathsf{el}$, merely composing has the effect of an existential query. This can be formalized in any equipment $\mathbb{D}$ by first viewing S as an extension and then composition with R on the left as in

$$
\begin{tikzcd}
 & S \arrow[r, "\mathrm{id}_S", "|" marking] \arrow[d, tail, "s"'] \arrow[dr, phantom, "\text{ext}"] & S \arrow[d, "!"] \\
X \arrow[r, "R"', "|" marking] & Y \arrow[r, "S"', "|" marking] & 1
\end{tikzcd}.
$$

In $\mathbb{R}\mathsf{el}$, the result $R \otimes S$ is precisely $\Diamond S$ as above. An equivalent $\mathbb{R}\mathsf{el}$-construction is to form the pullback on the left followed by the image

$$
\begin{tikzcd}
R \odot s^* \arrow[r, tail] \arrow[d, tail] \arrow[dr, phantom, "\lrcorner", very near start] & R \arrow[d, tail] \\
X \times S \arrow[r, tail, "1\times s"'] & X \times Y
\end{tikzcd}
\quad \rightsquigarrow \quad
\begin{tikzcd}
R \odot s^* \arrow[r, tail] \arrow[dr, two heads] & X \times S \arrow[r, "1\times !"] & X \times 1 \\
 & \Diamond S \arrow[ur, tail] &
\end{tikzcd}
$$

. That is, the corner of the pullback on the right is exactly the set of pairs (x,a) where $x \sim_R a$ under R. The image then binds $x \in X$ via existential quantification. This is formalized in any equipment $\mathbb{D}$ by first taking a restriction and then an extension:

$$
\begin{tikzcd}
X \arrow[r, "R\odot u^*", "|" marking] \arrow[d, "1"'] \arrow[dr, phantom, "\text{res}"] & S \arrow[d, tail, "s"] \\
X \arrow[r, "R"', "|" marking] & X
\end{tikzcd}
\quad \rightsquigarrow \quad
\begin{tikzcd}
X \arrow[r, "R\odot u^*", "|" marking] \arrow[d, "1"'] \arrow[dr, phantom, "\text{ext}"] & S \arrow[d, "!"] \\
X \arrow[r, "\Diamond S"', "|" marking] & 1
\end{tikzcd}.
$$

Now, in any equipment $\mathbb{D}$ there is a comparison cell of the form

$$
\begin{tikzcd}
X \arrow[r, "R\odot s^*", "|" marking] \arrow[d, "1"'] \arrow[dr, phantom, "\text{res}"] & S \arrow[r, "\mathrm{id}", "|" marking] \arrow[d, "s"] \arrow[dr, phantom, "\text{ext}"] & S \arrow[d, "!"] \\
X \arrow[r, "R"', "|" marking] & X \arrow[r, "S"', "|" marking] & 1
\end{tikzcd}
$$

Thus, there is a globular cell $\Diamond S \Rightarrow R \odot S$ since $\Diamond S$ is constructed as an extension:

$$
\begin{tikzcd}
X \arrow[rr, "R\odot s^*", "\shortmid"{marking}] \arrow[d] \arrow[drr, phantom, "\cong"] & & S \arrow[d] \\
X \arrow[r, "R\odot s^*", "\shortmid"{marking}] \arrow[d, "1"'] \arrow[dr, phantom, "\text{res}"] & S \arrow[r, "\mathrm{id}", "\shortmid"{marking}] \arrow[d, "s"] \arrow[dr, phantom, "\text{ext}"] & S \arrow[d, "!"] \\
X \arrow[r, "R"', "\shortmid"{marking}] & X \arrow[r, "S"', "\shortmid"{marking}] & 1
\end{tikzcd}
\quad = \quad
\begin{tikzcd}
X \arrow[r, "R\odot u^*", "\shortmid"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\text{ext}"] & S \arrow[d, "!"] \\
X \arrow[r, "\Diamond S"', "\shortmid"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\exists !"] & 1 \arrow[d, equal] \\
X \arrow[r, "R\odot S"', "\shortmid"{marking}] & 1
\end{tikzcd}
\tag{6.1}
$$

Now, it can be seen directly that this is an invertible cell when $\mathbb{D}$ is either $\mathbb{R}\mathsf{el}$ or $\mathbb{S}\mathsf{pan}$ or $\mathbb{M}\mathsf{at}(\mathsf{V})$ for suitably structured monoidal V. The case of $\mathbb{R}\mathsf{el}$ is trivial, so we treat the other two.

Example 6.1. In $\mathbb{S}\mathsf{pan}$, work with $\langle d, c\rangle \colon R \to X \times Y$ and then on the one hand

$$\Diamond S = \{((x,a),r) \mid x \in X, a \in S, r \in R, d(r) = x, c(r) = a\}$$

whereas $R \odot \hat{s}$ is the corner of the pullback:

$$R \times_X S = \{(r,a) \mid r \in R, a \in S, c(r) = a\}.$$

These are evidently isomorphic, as the latter just forgets the indices $x \in X$. But these are recoverable as $d(r) = x$. $\square$

Example 6.2. Let V denote any monoidal category with coproducts preserved by the tensor. We then have the double category $\mathbb{M}\mathsf{at}(\mathsf{V})$ of V-matrices (Example 5.1). To form the possibility operator, we have first the restriction

$$
\begin{tikzcd}
X \arrow[r, "M\odot s^*", "\shortmid"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\text{res}"] & S \arrow[d, "s"] \\
X \arrow[r, "M"', "\shortmid"{marking}] & X
\end{tikzcd}
\qquad (M \odot s^*)(x,a) = M(x,a) \qquad \text{for } x \in X, a \in S
$$

and then via an extension we have the possibility operator:

$$
\begin{tikzcd}
X \arrow[r, "M\odot s^*", "\shortmid"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\text{ext}"] & S \arrow[d, "!"] \\
X \arrow[r, "\Diamond S"', "\shortmid"{marking}] & !
\end{tikzcd}
\qquad \Diamond S(x) = \sum_{a \in S} M(x,a) \qquad \text{for } x \in X.
$$

Now, to compare this to the composite $M \odot S$ note first that S viewed as a proarrow $S \colon X \nrightarrow 1$ has the entries described by

$$
\begin{tikzcd}
S \arrow[r, "\mathrm{id}_S", "\shortmid"{marking}] \arrow[d, "s"'] \arrow[dr, phantom, "\text{ext}"] & S \arrow[d, "!"] \\
X \arrow[r, "S"', "\shortmid"{marking}] & 1
\end{tikzcd}
\qquad S(x) = \begin{cases} I & x \in S \\ 0 & \text{o/w} \end{cases} \qquad \text{for } x \in X.
$$

This follows from the definition of the unit proarrows in $\mathbb{M}\mathsf{at}(\mathsf{V})$. Now, from the definition of the loose composite of two matrices, we have

$$\begin{aligned}(M \odot S)(x) &= \sum_{y\in X} M(x,y) \otimes S(y)\\ &= \sum_{y\in X} M(x,y) \otimes \begin{cases} I & y \in U \\ 0 & \text{o/w}\end{cases}\\ &= \sum_{y\in S} M(x,y)\end{aligned}$$

for any $x \in X$, showing that the two constructions are the same in this case too. □

Now, in fact the two construction agree in general for any equipment such as $\mathbb{P}\mathsf{rof}$ or $\mathbb{M}\mathsf{at}(\mathsf{V})$ for more complicated V, each of which are difficult to treat directly.

Proposition 6.3. *The globular cell $\Diamond S \Rightarrow R \odot S$ induced in Equation* (6.1) *is an isomorphism in any equipment. Accordingly, the two possible definitions of $\Diamond S$ agree.*

Proof. It suffices to show that the two sides of Equation (6.1) compute the same extension. But this is just a result of how they are computed in equipments. That is, S viewed as a proarrow defined via the extension is the composite $S := s^* \odot \mathrm{id}_S \odot !_S$ where $!_S$ is the companion of $!\colon S \to 1$. So, the composite is $R \odot S = R \odot s^* \odot \mathrm{id}_S \odot !_S$. Likewise, $\Diamond S$ as the extension of $R \odot s^*$ along 1_X and $!$ is just $\Diamond S = R \odot s^* \odot !_S$. These are clearly isomorphic by inserting appropriate associators and units coming from the equipment structure. More abstractly, the left side of Equation (6.1) can thus be rearranged to make $R \odot S$ exhibit the same extension explicitly. □

So, although the constructions are thus essentially the same, we make a choice as to which is basic. We insist on starting with the subtype $s\colon S \to Y$ consistent with the set-up of Section 3. And so in either case there are two operations to perform the query. There is a slight semantic advantage to the latter. For in the case of $\mathbb{S}\mathsf{pan}$, although the appropriate element $x \in X$ is implicit in the composite as $d(r) = x$, our preference is to keep the data as a triple $((x,a),r) = (x,a,r)$ as in the second way of computing the possibility operator. This way all the information is recorded in the apex of the span and the two legs can be forgotten.

Definition 6.4 (Possibility)**.** Let $s\colon S \rightarrowtail X$ denote any monic arrow in the equipment $\mathbb{D}$. The proposition **possibly** S is then formed as an restriction followed by an extension:

$$\begin{tikzcd} X \arrow[r, "R\odot u^*", "\shortmid"{marking}] \arrow[d, "1"'] \arrow[dr, phantom, "\text{res}"] & S \arrow[d, tail, "s"] \\ X \arrow[r, "R"', "\shortmid"{marking}] & X \end{tikzcd} \quad \rightsquigarrow \quad \begin{tikzcd} X \arrow[r, "R\odot u^*", "\shortmid"{marking}] \arrow[d, "1"'] \arrow[dr, phantom, "\text{ext}"] & S \arrow[d, "!"] \\ X \arrow[r, "\Diamond S"', "\shortmid"{marking}] & 1 \end{tikzcd}$$

as in the pattern suggested above: □

Note then that the content of Proposition 6.3 is thus that of a *rewrite rule* for the possibility operator and thus also for existential queries of this kind.

All the instances of existential quantification introduced are instances of *fibrational semantics* of the equipment $\mathbb{D}$. We have the general picture:

$$\Sigma_{f,g}\colon \mathbb{D}(A,B) \rightleftarrows \mathbb{D}(X,Y)\colon (f,g)^* \qquad \Sigma_{f,g} \dashv (f,g)^* \tag{6.2}$$

for any arrows $f\colon A \to X$ and $g\colon B \to Y$ of $\mathbb{D}$. The hom categories are the fibers of $\langle \mathrm{src}, \mathrm{tgt}\rangle\colon \mathbb{D}_1 \to \mathbb{D}_0 \times \mathbb{D}_0$ over the pairs of objects (A, B) and (X, Y). Here $\Sigma_{f,g}$ denotes the left adjoint to substitution/restriction along f and g. This is provided automatically since $\langle \mathrm{src}, \mathrm{tgt}\rangle$ is both a fibration and opfibration [Shu08]. Of course $\Sigma_{f,g}$ is thus extension in the equipment structure. Now, again, both instances of existential quantification are recoverable in this framework. On the one hand, for $s\colon S \to Y$ we have the functors

$$\mathbb{D}(X, Y) \xrightarrow{(1_X,s)^*} \mathbb{D}(X, S) \xrightarrow{\Sigma_S} \mathbb{D}(X, 1)$$

and consequently

$$\Sigma_S(1_X, s)^*(R) = \Diamond S$$

as in Definition 6.4. On the other hand, ordinary existential quantification is just an application of a left adjoint:

$$\mathbb{D}(X, Y) \xrightarrow{\Sigma_X} \mathbb{D}(1, Y) \qquad \Sigma_X(R) = \exists x.R(x, y)$$

as has already been observed in the case of $\mathbb{R}\mathsf{el}$. In each case, the left adjoints used are special cases of *Kan quantification.* Let $f\colon A \to X$ denote any morphism of $\mathbb{D}$. Consider the *one-sided* adjunction

$$\Sigma_f\colon \mathbb{D}(A, Y) \rightleftarrows \mathbb{D}(X, Y)\colon f^* \qquad \Sigma_f \dashv f^* \tag{6.3}$$

Owning to the nature of the computation of restrictions and extensions, the two functors are simply: *precompose with a companion or conjoint* as in $\Sigma_f = f^* \odot -$ and $f^* = f_! \odot -$ and the adjunction is then

$$f^* \odot -\colon \mathbb{D}(A, Y) \rightleftarrows \mathbb{D}(X, Y)\colon f_! \odot - \qquad f^* \odot - \dashv f_! \odot - \tag{6.4}$$

Retaining the Σ-notation for the left adjoint, the adjunction expresses an existential quantification introduction rule. Namely, that there is a bijection between globular cells

$$\begin{tikzcd} X \arrow[r, "\Sigma_f R", "\shortmid"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\Downarrow"] & Y \arrow[d, equal] \\ X \arrow[r, "S"', "\shortmid"{marking}] & Y \end{tikzcd} \quad \leftrightarrow \quad \begin{tikzcd} A \arrow[rr, "R", "\shortmid"{marking}] \arrow[d, equal] & \arrow[d, phantom, "\Downarrow"] & Y \arrow[d, equal] \\ A \arrow[r, "f_!"', "\shortmid"{marking}] & X \arrow[r, "S"', "\shortmid"{marking}] & Y \end{tikzcd} \qquad \frac{\Sigma_f R \Rightarrow S}{R \Rightarrow f_! \odot S}$$

With the computation $\Sigma_f = f^* \odot -$ we have an analogue of the computation of the left adjoint to substitution for V-functors in [Law02, §3]. Applying the bijection to the identity on $\Sigma_f R$, there is a single cell

$$\begin{tikzcd} A \arrow[rr, "R", "\shortmid"{marking}] \arrow[d, equal] & \arrow[d, phantom, "\Downarrow"] & Y \arrow[d, equal] \\ A \arrow[r, "f_!"', "\shortmid"{marking}] & X \arrow[r, "\Sigma_f R"', "\shortmid"{marking}] & Y \end{tikzcd}$$

and the universal property makes $\Sigma_f R$ into a (left) *Kan extension* of R along $f_!$ in the underlying loose bicategory of $\mathbb{D}$. Specializing to the case of Σ_X as above and writing the traditional '$\exists$' for the left adjoint, the bijection expressing the universal property is precisely the usual existential introduction rule:

$$\begin{tikzcd} X \arrow[r, "\exists_X R", "\shortmid"{marking}] \arrow[d, "!"'] \arrow[dr, phantom, "\Downarrow"] & Y \arrow[d, equal] \\ 1 \arrow[r, "S"', "\shortmid"{marking}] & Y \end{tikzcd} \quad \leftrightarrow \quad \begin{tikzcd} X \arrow[rr, "R", "\shortmid"{marking}] \arrow[d, equal] & \arrow[d, phantom, "\Downarrow"] & Y \arrow[d, equal] \\ X \arrow[r, "X_!"', "\shortmid"{marking}] & 1 \arrow[r, "S"', "\shortmid"{marking}] & Y \end{tikzcd} \qquad \frac{\exists_X R \Rightarrow S}{R \Rightarrow X_! \odot S}$$

Notice that since this construction provides a left Kan extension in the underlying bicategory of $\mathbb{D}$ and thus that $\mathbb{D}$ is in a sense bicategorically closed, this is not a fragment of the closed structure studied in [Shu08].

7 Universal Queries

Now that we have made good sense of existential quantification, we will turn to universal quantification, starting from the fibrational semantics and Kan quantification. That the restriction functor $(f,g)^*\colon \mathbb{D}(x,y) \to \mathbb{D}(a,b)$ has a right adjoint is the statement that there exists a functor

$$\prod_{f,g}\colon \mathbb{D}(a,b) \to \mathbb{D}(x,y)$$

and natural isomorphisms

$$\mathbb{D}(a,b)(f_! m g^*, s) \cong \mathbb{D}(x,y)(m, \prod_{f,g}(s))$$

establishing natural families of bijections of globular cells

$$\frac{f_! m g^* \Rightarrow s}{m \Rightarrow \prod_{f,g}(s)}$$

given in each direction by application of the appropriate functor and then composition with a unit or counit as the case may be. The elementary characterization of this right adjoint as a universal is the following.

Proposition 7.1. *The restriction functor $(f,g)^*\colon \mathbb{D}(x,y) \to \mathbb{D}(a,b)$ has a right adjoint if, and only if, for each $s\colon a \nrightarrow b$ there is a globular cell ϵ_s of the form*

$$\begin{tikzcd} a \arrow[r, "f_!(\forall_{f,g}(s))g^*", "\shortmid"{marking}] \arrow[d, "f"'] \arrow[dr, phantom, "res"] & b \arrow[d, "g"] \\ x \arrow[r, "\forall_{f,g}(s)"', "\shortmid"{marking}] & y \end{tikzcd} \quad \rightsquigarrow \quad \begin{tikzcd} a \arrow[r, "f_!(\forall_{f,g}(s))g^*", "\shortmid"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\epsilon_s"] & b \arrow[d, equal] \\ a \arrow[r, "s"', "\shortmid"{marking}] & b \end{tikzcd}$$

such that for any other globular cell from a restriction along f and g, say, of the form

$$\begin{tikzcd} a \arrow[r, "f_! w g^*", "\shortmid"{marking}] \arrow[d, "f"'] \arrow[dr, phantom, "res"] & b \arrow[d, "g"] \\ x \arrow[r, "w"', "\shortmid"{marking}] & y \end{tikzcd} \quad \rightsquigarrow \quad \begin{tikzcd} a \arrow[r, "f_! w g^*", "\shortmid"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\xi"] & b \arrow[d, equal] \\ a \arrow[r, "s"', "\shortmid"{marking}] & b \end{tikzcd}$$

there is a unique cell $\hat{\xi}$ for which h factors through ϵ_s as in

$$\begin{tikzcd} a \arrow[r, "f_! w g^*", "\shortmid"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\hat{\xi}"] & b \arrow[d, equal] \\ a \arrow[r, "f_!(\forall_{f,g}(s))g^*", "\shortmid"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\epsilon_s"] & b \arrow[d, equal] \\ a \arrow[r, "s"', "\shortmid"{marking}] & b \end{tikzcd} \quad = \quad \begin{tikzcd} a \arrow[r, "f_! w g^*", "\shortmid"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\xi"] & b \arrow[d, equal] \\ a \arrow[r, "s"', "\shortmid"{marking}] & b \end{tikzcd}$$

Proof. This is just the characterization of an adjunction in terms of the existence of a counit as a universal ([Mac98, §IV.1, Theorem 2(iv)]). Note that naturality of ϵ_s in s follows automatically. □

8 Tabulators and Modality

9 The Role of Compactness

10 Structure on Relations

This operator justly bears the '$\Diamond$' notation not only on account of the semantics discussed above. There is also the following observation, namely, that what is the case is possibly the case, as in the usual monadic interpretation of possibility. The proof is most naturally interpreted in $\mathbb{R}\mathsf{el}$, but is completely formal. When $\mathbb{R}\mathsf{el}$ is meant, the relationship is the familiar $S \leq \Diamond S$.

Proposition 10.1. *If R is reflexive, then there is a globular cell $S \Rightarrow \Diamond S$.*

Proof. Reflexivity of R means that there is a canonical cell

$$\begin{tikzcd}
X \arrow[r, "\mathrm{id}_X", "\shortmid"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\delta_X"] & X \arrow[d, equal] \\
X \arrow[r, "R"', "\shortmid"{marking}] & X
\end{tikzcd}$$

expressing the fact that the diagonal factors through R. Consequently, since $R \odot u^*$ arises as a restriction, there is a unique cell making an equality

$$\begin{tikzcd}
S \arrow[r, "\mathrm{id}_S", "\shortmid"{marking}] \arrow[d, "s"'] \arrow[dr, phantom, "\mathrm{id}_s"] & S \arrow[d, "s"] \\
X \arrow[r, "\mathrm{id}_X", "\shortmid"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\delta_X"] & X \arrow[d, equal] \\
X \arrow[r, "R"', "\shortmid"{marking}] & X
\end{tikzcd}
\quad = \quad
\begin{tikzcd}
S \arrow[r, "\mathrm{id}_S", "\shortmid"{marking}] \arrow[d, "s"'] \arrow[dr, phantom, "\exists !"] & S \arrow[d, equal] \\
X \arrow[r, "R\odot s^*", "\shortmid"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\mathrm{res}"] & S \arrow[d, "s"] \\
X \arrow[r, "R"', "\shortmid"{marking}] & X
\end{tikzcd}$$

And so, putting this cell as (I) in the diagram below, we get the desired comparison induced by the universal property of the extension cell:

$$\begin{tikzcd}
S \arrow[r, "\mathrm{id}_S", "\shortmid"{marking}] \arrow[d, "s"'] \arrow[dr, phantom, "(\mathrm{I})"] & S \arrow[d, equal] \\
X \arrow[r, "R\odot s^*", "\shortmid"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\mathrm{ext}"] & S \arrow[d, "!"] \\
X \arrow[r, "\Diamond S"', "\shortmid"{marking}] & 1
\end{tikzcd}
\quad = \quad
\begin{tikzcd}
S \arrow[r, "\mathrm{id}_S", "\shortmid"{marking}] \arrow[d, "s"'] \arrow[dr, phantom, "\mathrm{ext}"] & S \arrow[d, "!"] \\
X \arrow[r, "S", "\shortmid"{marking}] \arrow[d, equal] \arrow[dr, phantom, "\exists !"] & 1 \arrow[d, equal] \\
X \arrow[r, "\Diamond S"', "\shortmid"{marking}] & 1
\end{tikzcd}$$

Notice the abuse of notation, namely, that we think of S as both the object providing the domain of s but also as a proarrow $S\colon X \nrightarrow 1$ obtained by extension. $\square$

11 Full Closed Comprehension

12 Prospectus

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